

Notes on Estimation of GBM and mean reverting processes

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1 Mean reverting process

Consider a mean reverting process for cash rate $r(t)$:

$$dr(t) = \theta(\gamma - r(t))dt + \sigma dW(t), \quad (1)$$

where $W(t)$ is Brownian motion, σ is volatility, γ is the mean reverting level and θ is the mean reversion speed. The well-known solution of this process for discrete time with equal time step $\Delta t = t_{i+1} - t_i$ is

$$r(t_{i+1}) = r(t_i) \times w + \gamma \times (1 - w) + v \times \epsilon_{i+1}, i = 0, 1, \dots$$

where

$$w = e^{-\theta \times \Delta t} \text{ and } v = \sqrt{\frac{\sigma^2}{2\theta}(1 - w^2)}$$

and $\epsilon_1, \epsilon_2, \dots$ are independent and identically distributed random variables form standard normal distribution $N(0, 1)$.

Using data $r(t_0), r(t_1), \dots, r(t_N)$, parameters (γ, θ, σ) can be estimated by maximizing the likelihood of data (conditional on r_0)

$$L = \prod_{i=1}^N \frac{1}{\sqrt{2\pi}v} e^{-\frac{1}{2}\tilde{\epsilon}_i^2/v^2}, \quad \tilde{\epsilon}_i = r(t_i) - wr(t_{i-1}) - \kappa$$

equivalent to minimizing

$$\ell = \frac{1}{2v^2} \sum_{i=0}^{N-1} (r(t_{i+1}) - r(t_i) \times w - \gamma(1 - w))^2 + N \ln(v)$$

with respect to w , $\kappa = \gamma(1 - w)$, and v . Solving the first order conditions gives as follows:

$$\frac{\partial}{\partial v} \ell = -\frac{1}{v^3} \sum_{i=1}^N \epsilon_i^2 + N \frac{1}{v} = 0 \Rightarrow v^2 = \frac{1}{N} \sum_{i=1}^N \epsilon_i^2, \quad (2)$$

$$\frac{\partial}{\partial \kappa} \ell = 0 \Rightarrow w \sum_{i=0}^{N-1} r(t_i) + \kappa N = \sum_{i=0}^{N-1} r(t_{i+1}), \quad (3)$$

$$\frac{\partial}{\partial w} \ell = 0 \Rightarrow w \sum_{i=0}^{N-1} r(t_i)r(t_i) + \kappa \sum_{i=0}^{N-1} r(t_i) = w \sum_{i=0}^{N-1} r(t_i)r(t_{i+1}). \quad (4)$$

Solving these gives

$$\hat{w} = \frac{1}{d} \left[\sum_{i=0}^{N-1} r(t_{i+1}) \sum_{i=0}^{N-1} r(t_i) - N \sum_{i=0}^{N-1} r(t_i) r(t_{i+1}) \right], \quad (5)$$

$$\hat{\kappa} = \frac{1}{d} \left[\sum_{i=0}^{N-1} r(t_i) \sum_{i=0}^{N-1} r(t_i) r(t_{i+1}) - \sum_{i=0}^{N-1} r(t_i) r(t_i) \sum_{i=0}^{N-1} r(t_{i+1}) \right], \quad (6)$$

where

$$d = \sum_{i=0}^{N-1} r(t_i) \sum_{i=0}^{N-1} r(t_i) - N \sum_{i=0}^{N-1} r(t_i) r(t_i).$$

Then

$$\hat{v}^2 = \frac{1}{N} \sum_{i=1}^N \hat{\epsilon}_i^2, \quad \hat{\epsilon}_i = r(t_i) - \hat{w}r(t_{i-1}) - \hat{\kappa}.$$

Finally, the original parameters are estimated as

$$\hat{\theta} = \frac{\ln \hat{w}}{\Delta t}, \quad (7)$$

$$\hat{\gamma} = \frac{\hat{\kappa}}{1 - \hat{w}}, \quad (8)$$

$$\hat{\sigma}^2 = 2 \frac{\hat{\theta} \hat{v}^2}{1 - \hat{w}^2}, \quad (9)$$

and model residuals are

$$\hat{\epsilon}_i = \frac{r(t_i) - \hat{w}r(t_{i-1}) - \hat{\kappa}}{\hat{v}}.$$

2 GBM process for S&P500

Consider GBM process for S&P500

$$dS(t)/S(t) = \mu dt + \sigma_S dW_S(t).$$

where μ is the drift, σ_S is volatility and $W_S(t)$ is Brownian motion. The well-known solution of this process for discrete time with equal time step is

$$S(t_{i+1}) = S(t_i) e^{(\mu - \frac{1}{2}\sigma_S^2)\Delta t + \sigma_S \sqrt{\Delta t} \epsilon_{i+1}^S}, \quad i = 0, 1, \dots,$$

where $\epsilon_1^S, \epsilon_2^S, \dots$ are independent and identically distributed random variables form the standard normal distribution $N(0, 1)$. Here, ϵ_i^S can be correlated to disturbance ϵ_i in mean reverting process for cash rate, with linear correlation coefficient

$$\rho = E[\epsilon_i \epsilon_i^S].$$

Model parameters of GBM process can be estimated using data $S(t_0), \dots, S(t_N)$ via the maximum likelihood method resulting in

$$\hat{\mu} = \frac{1}{N} \sum_{i=0}^{N-1} \ln \frac{S(t_{i+1})}{S(t_i)}, \quad (10)$$

$$\hat{\sigma}_S^2 = \frac{1}{N} \sum_{i=0}^{N-1} \left(\ln \frac{S(t_{i+1})}{S(t_i)} - \hat{\mu} \right)^2, \quad (11)$$

$$\hat{\mu} = \hat{\mu} + \frac{1}{2} \hat{\sigma}_S^2, \quad (12)$$

and model residuals are

$$\hat{\epsilon}_i^S = \left(\ln \frac{S(t_{i+1})}{S(t_i)} - \hat{\mu} \right) \times \frac{1}{\hat{\sigma}_S}.$$

Finally, correlation between S&P500 and cash rate disturbances is estimated using residuals as

$$\hat{\rho} = \sum_{i=1}^N \hat{\epsilon}_i \hat{\epsilon}_i^S.$$

3 Mean reverting process for S&P500

Consider the following process for S&P500 with mean reverting to the expected mean with annual time step

$$M(t_{i+1}) = M(t_i)(1 + \mu_B), \quad (13)$$

$$S(t_{i+1}) = S(t_i)(1 + \mu_B + \sigma_B \epsilon_{i+1}^B) + \gamma_B(M(t_i) - S(t_i)), \quad (14)$$

where $M(t_0) = S(t_0)$, μ_B is the drift, σ_B is volatility, γ_B is mean reversion parameter, and $\epsilon_1^B, \epsilon_2^B, \dots$ are independent and identically distributed random variables from the standard normal distribution $N(0, 1)$. Here, ϵ_i^B can be correlated to disturbance ϵ_i in mean reverting process for cash rate, with linear correlation coefficient

$$\rho_B = E[\epsilon_i \epsilon_i^B].$$

Model parameters of this process can be estimated using data $S(t_0), \dots, S(t_N)$ via the maximum likelihood method resulting in

$$L = \prod_{i=1}^N \frac{1}{\sqrt{2\pi}\sigma_B} e^{-\frac{1}{2}(\tilde{\epsilon}_i^B)^2/\sigma_B^2}, \quad \tilde{\epsilon}_{i+1}^B = \frac{S(t_{i+1}) - \gamma_B(M(t_i) - S(t_i))}{S(t_i)} - 1 - \mu_B$$

equivalent to minimizing

$$\ell = \frac{1}{2\sigma_B^2} \sum_{i=1}^N (\tilde{\epsilon}_i^B)^2 + N \ln(\sigma_B)$$

with respect to $\sigma_B, \gamma_B, \mu_B$. It simplifies to minimizing $\sum_{i=1}^N (\tilde{\epsilon}_i^B)^2$ with respect to μ_B and γ_B to find $\hat{\mu}_B$ and $\hat{\gamma}_B$. Then

$$\hat{\sigma}_B^2 = \frac{1}{N} \sum_{i=1}^N \left(\hat{\tilde{\epsilon}}_i^B \right)^2,$$

and model residuals are

$$\hat{\tilde{\epsilon}}_{i+1}^B = \left(\frac{S(t_{i+1}) - \hat{\gamma}_B(M(t_i) - S(t_i))}{S(t_i)} - 1 - \hat{\mu}_B \right) \times \frac{1}{\hat{\sigma}_B}.$$

Finally, correlation between S&P500 disturbance ϵ_i^B and cash rate disturbance ϵ_i is estimated using residuals as

$$\hat{\rho} = \sum_{i=1}^N \hat{\epsilon}_i \hat{\epsilon}_i^B.$$